CD Engine Shop Receives the 4th Consecutive FAA Diamond Award

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Engine Maintenance Training (EMT)

FAA Approved (8 HR) IA - Refresher Training Program

Presented by

Mike Straus
CD Aviation Services

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NOTE this publication has no official status, in all cases refer to the FAA authorized documents ie, airframe FM, MM and engine MM for approved data.

Service Programs

Turboprop

- TPE331-1 through -12 series

Turbofan

- TFE731-2, -3, -4, -5, -15, -20, -40, -50, -60 series

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Auxiliary Power Units

- GTCP 36 -100 and 150 series

TPE 331 History

The TPE 331 engine was first manufactured in 1965 and they were certified at 575 HP.

Those first engines weighed 335 lbs which give a HP to weight ratio of about 1.7 HP per lb.

Several upgraded models have been developed since 1965 but they all have the same external dimensions.

The engine weight has increased slightly over the years but the HP has nearly doubled to 1100 HP.

As a result the HP to weight ratio has improved to 2.7 HP per lb. which is a 59% improvement.

Since 1965 over 13,000 TPE331 engines have been installed on over 85 different AC models and in excess of 113 million engine flight hours has been logged.

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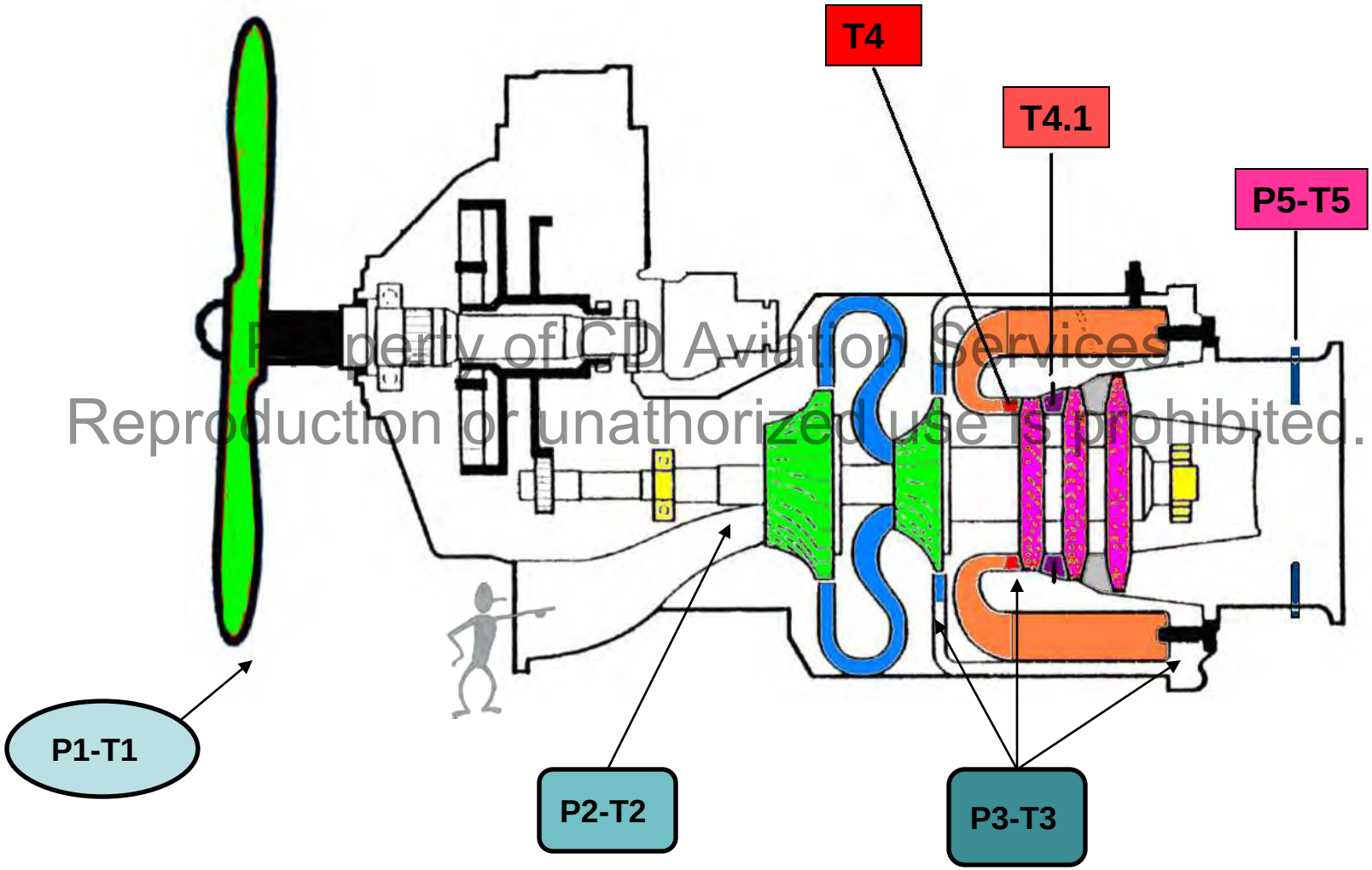
Air flow stations continued



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BORESCOPE VIEW OF
1ST TURBINE WHEEL BLADES
(STATION T 4)

Horsepower

Model	std day HP rating	Thermo dynamic SHP	OAT @ max	SFC	PROP	SPEED
-1	665	715	75'f	0.605	CW	2000
-2	715	715	59'f	0.558	CW	2000
-3	840	840	59'f	0.590	CW	2000
-5	776	840	86'f	0.590	CCW	1591
-6	715	840	92'f	0.590	CW	2000
-8	715	865	97'f	0.568	CW	2000
-10	900	1000	91'f	0.558	CCW	1591
-11	1000	1000	59'f	0.558	CCW	1591
-12	1050	1100	59'f	0.547	CCW	1591

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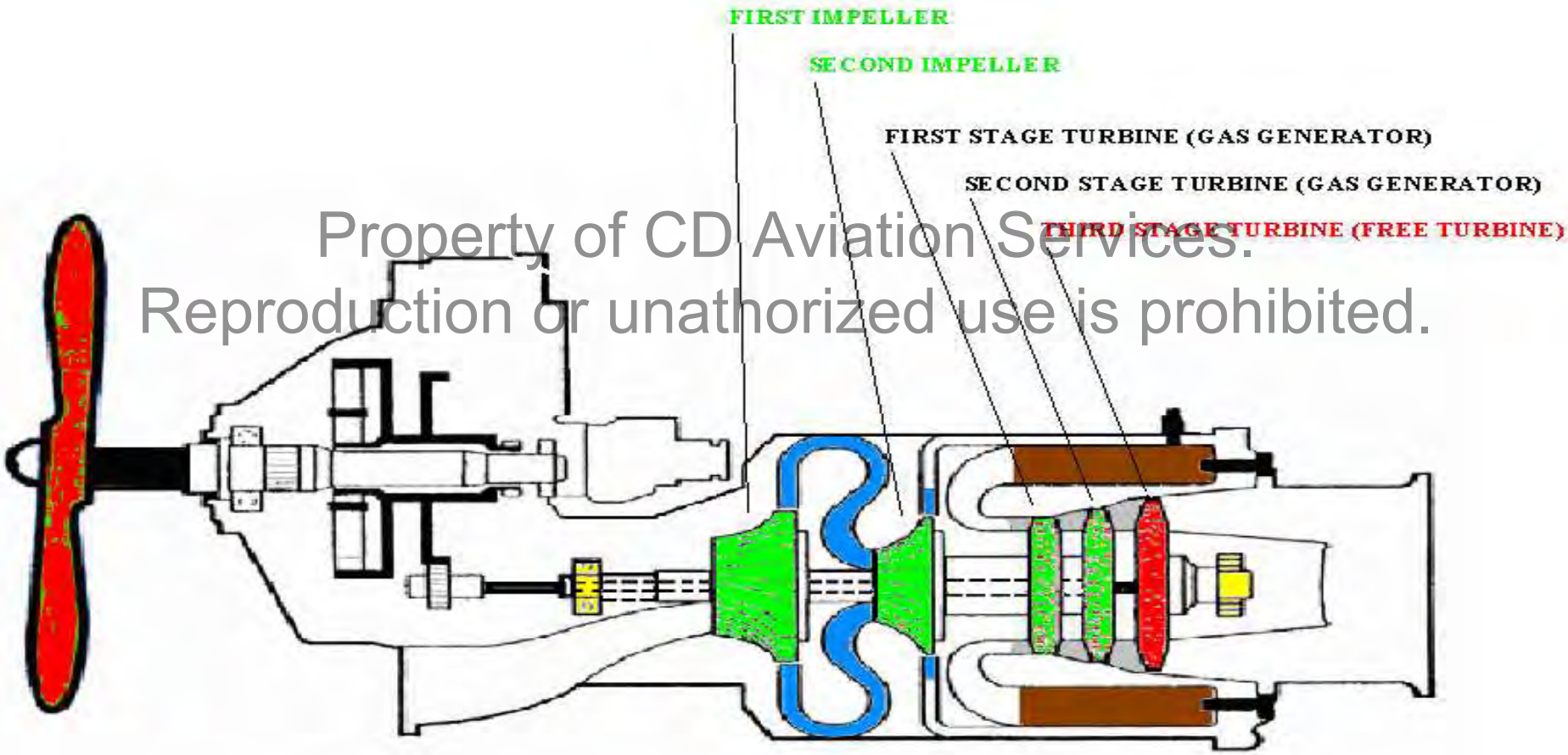
Free Turbine Design (DRAWING ONLY)

FREE OR POWER TURBINE DESIGN



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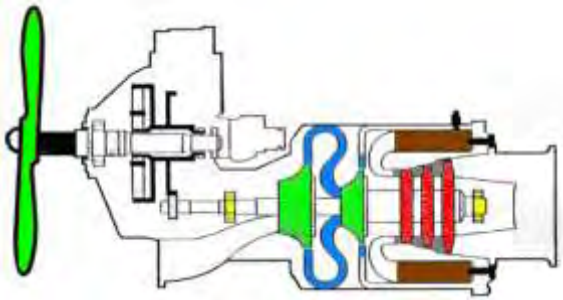
Free Turbine Design

Free turbine engine (PT6) advantages versus fixed shaft engine designs such as the TPE331.



The free turbine engine starts are faster and cooler with less reliance on a good start system and or start system energy.

On the free turbine design the starter energy available is not reduced by the amount of energy required to drive the propeller like it is on a fixed shaft engine. As a result all the starter energy is directed to start the free turbine design engine.

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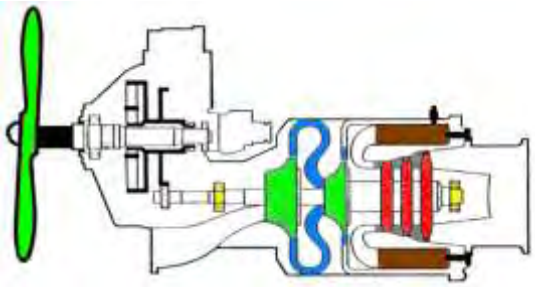
Bowed rotor, critical range issues and load from the propeller all contribute to the engine being more difficult to start and require full operator attention during starts to prevent potential engine damage on the fixed shaft design.

The free turbine design adapts to helicopter installations much better and they are quieter during ground handling operations.

Fixed shaft Design



Fixed shaft engine (TPE 331 design) advantages versus the free turbine design

The fixed shaft engine design is a more fuel efficient design that significantly reduces hourly fuel cost of operation.

The ability to provide power rapidly (without a spool up time - delay) is a significant safety advantage if the pilot gets a little behind the power curve.

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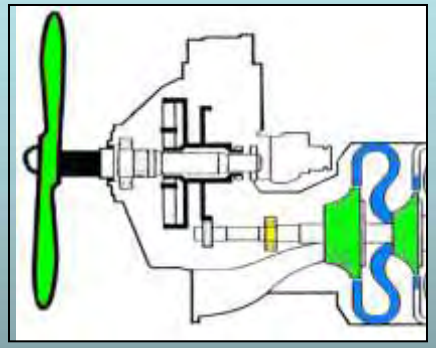
The overhaul cost on the TPE are significantly below the cost of the PT6.

The ram air inlet and the straight thru exhaust enhance engine and aircraft in-flight performance versus the free turbine configuration.

Constant Speed Operation *(fixed shaft)*

In a constant speed (fixed shaft engine) the propeller pitch or prop load plus the load from the compressor section together must be offset by the turbine section power output.



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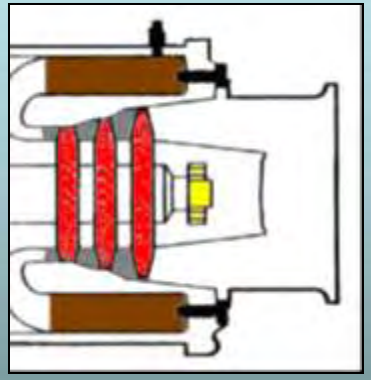
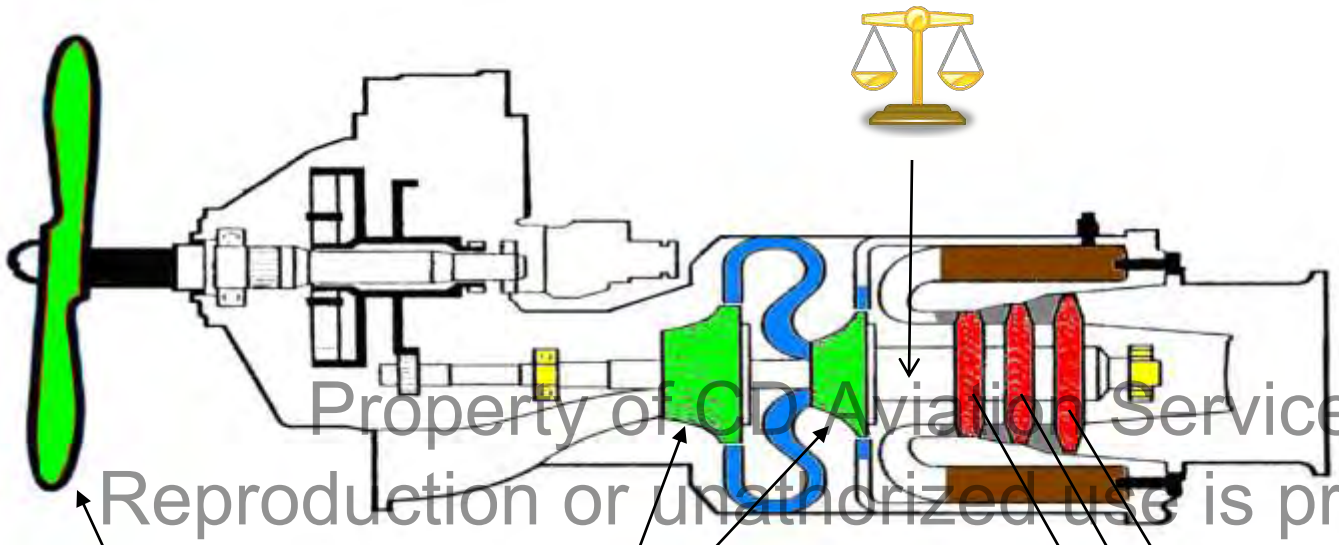
The balance between turbine section power output and *the load applied by the compressor section and the prop pitch* must be equal to result in a fixed power and speed setting.



Constant Speed Operation *(fixed shaft)*



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1000 HP

2000 HP

3000 HP

COMPRESSOR IMPELLERS AND PROP PITCH LOADS MUST!

=

TURBINE POWER OUTPUT!

Constant Speed Operation *(fixed shaft)*

Added fuel to the turbine combustion section increases the EGT / ITT and the pneumatic energy in the hot section. The turbines then increase in speed which momentarily unbalances the forces between the turbine output and the resisting loads of the compressor and the propeller pitch.

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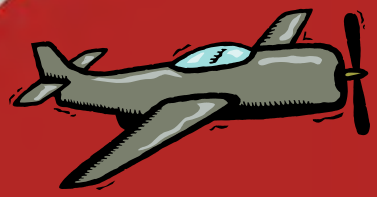
The Propeller governor flyweights sense this speed increase and automatically reduce the PG oil flow and pressure going to the prop which increases the prop pitch and the load until the system is back in balance and on speed.



Cost Comparisons

	commercial operator	
specification	PT6-114	331-12
oh interval hrs	7000	7000
hs interval hrs	1800	3500
hs events per oh	3	1
oh cost	$ 330,000	$ 200,000
hs cost (per oh cycle)	$ 99,000	$ 50,000
oh & hs cost per hr	$ 61.29	$ 35.71
higher cruise ff (PPH) at same speed	40	0
gallons of fuel per hr	6.45	
fuel cost per gallon	4	4
higher fuel cost per hr	$ 25.81	0
total hs + oh + fuel cost per hr	$ 87.09	$ 35.71
total cost each 5000 hr		
tpe savings on 5K program		
total cost each 7000 hr	$ 609,645	$ 250,000
tpe savings on 7K program		$ 359,645

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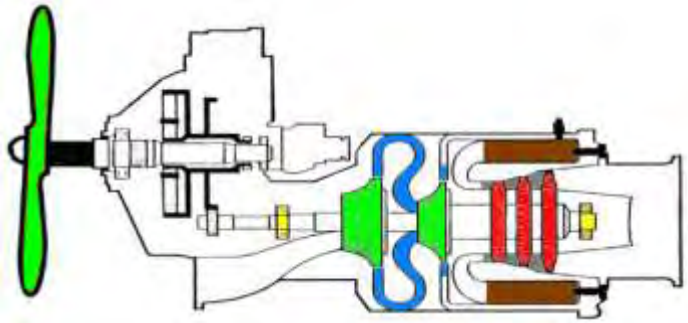
Engine Power



LIMITATIONS

The TPE 331 max allowed power limit is either torque or temperature (EGT or ITT)!

The operator advances the power levers as required without exceeding the temperature or torque limits!

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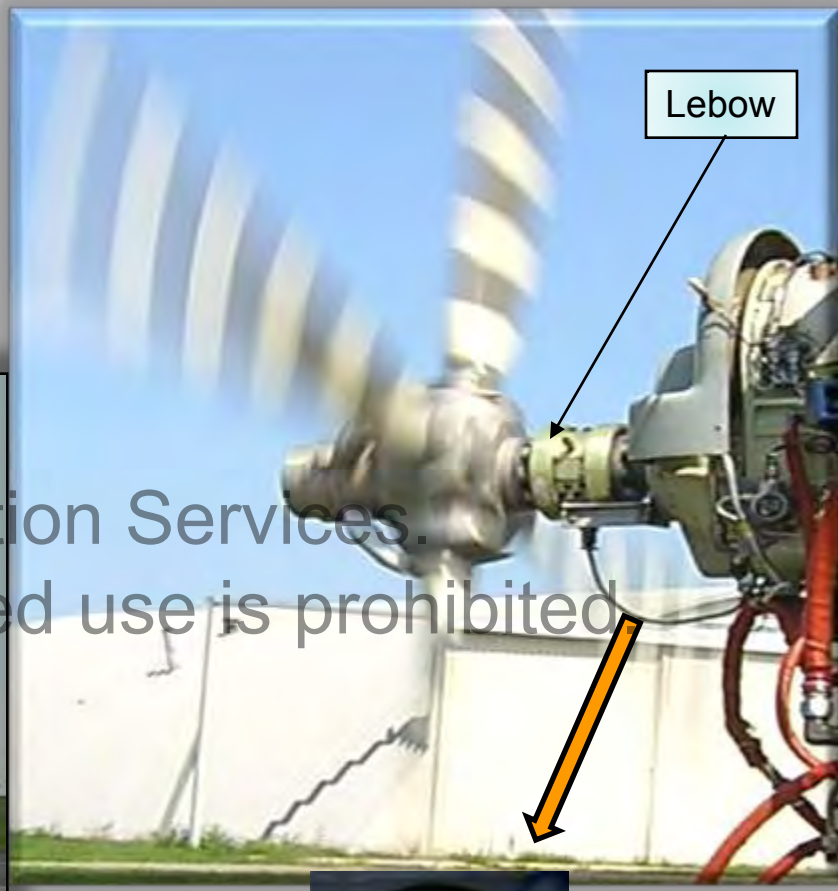
The turbine temperature limit is established by the engine manufacture and considers the turbine materials and the rotational speeds of the power section.

The torque limits are established by the airframe manufacture to meet the aircraft structural integrity and design performance criteria.

ENGINE ON (CDAS JOPLIN) TEST CELL

A lebow is used to determine what the real torque is at the prop via the lebow digital torque readout and what the torque sensor pressures positive pressures are on torque sensor engines. As well as what the torque ring voltages are on strain gage engines at these various lebow indicated power levels.



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ENGINE EGT/ITT SYSTEMS

Performance Testing

The test cell computer performance program computes the T4 value in relationship to EGT during engine test cell performance runs.

The TPE 331 design uses an EGT compensator that changes the raw EGT reading to a compensated EGT. When the compensator is set to min comp it reads the EGT limit when the engine is making rated power on a standard day and the T4 is below the limit.

A max compensation setting is also available so that the engine compensated EGT indication reads less than it does with the min comp setting. This allows the operator to advance the PL so that the T4 can be increased to the T4 limit at the same time the EGT reaches the red line using all the engines temperature margin.

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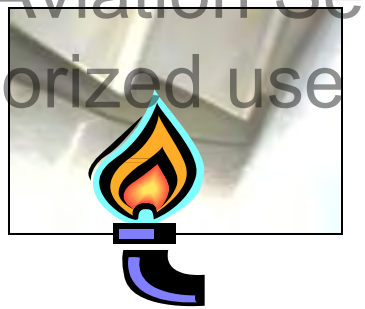
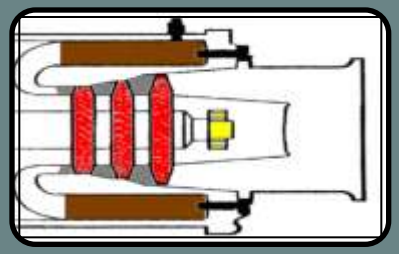


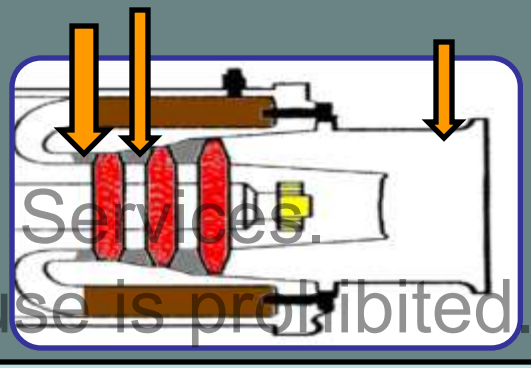
ENGINE EGT/ITT SYSTEMS

The T4 temperature location is at the inlet of the first stage turbine wheel and it is the most important and highest temperature in the turbine wheel section and is the area that limits the thermal power of the engine.

In both the ITT and EGT engines they must reach the engine temp gauge red line at or before the T4 reaches its limit to ensure that engine damage does not occur and that the engine life is not adversely effected.



The temperature drop from station T4 to station T4.1 (ITT location) is very linear with a specific temperature drop or change between these two locations that is not effected by the mass airflow through the engine.



However, the temperature drop between stations T4 and T5 (EGT) is affected by the mass air flow of the engine and is not linear.

As a result a standard EGT indication uncorrected and without a system such as the SRL system requires a constant reference to the flight manual to determine the EGT limit for the specific ambient conditions at any given point in time (which is burdensome for the operator).

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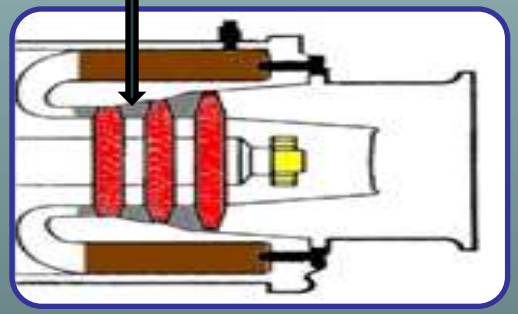
ENGINE EGT/ITT SYSTEMS

Station T4.1 (ITT harness location) is on the exit side of the first stage turbine wheel. Bolted into the vane openings of the 2nd stage nozzle or stator assembly on the TPE -3 -5 and -6 model engines.

T4.1 is a very hostile location for the harness due to the high temperatures as well as solids and carbon erosion that occurs at this location on these older model engines.

The ITT harness, primarily due to the location within the engine has not proven to be as reliable as the alternative EGT harness configuration. When the ITT harness fails prior to the schedule hot section it requires more labor and material to replace the harness because an engine removal and partial hot section disassembly is required.

In addition the ITT harness change results in exposure to other expensive hot section hardware in addition to the ITT harness that can effect the total cost of the repair.



The good news is the ITT system is a very responsive system that does not require a SRL system (computer) to provide a single red line indication with 923°c being the redline takeoff limit.

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ENGINE EGT/ITT SYSTEMS

EGT (T5) probes (8 each)

ITT (T4.1) Harness

EGT (T5) Harness

comp

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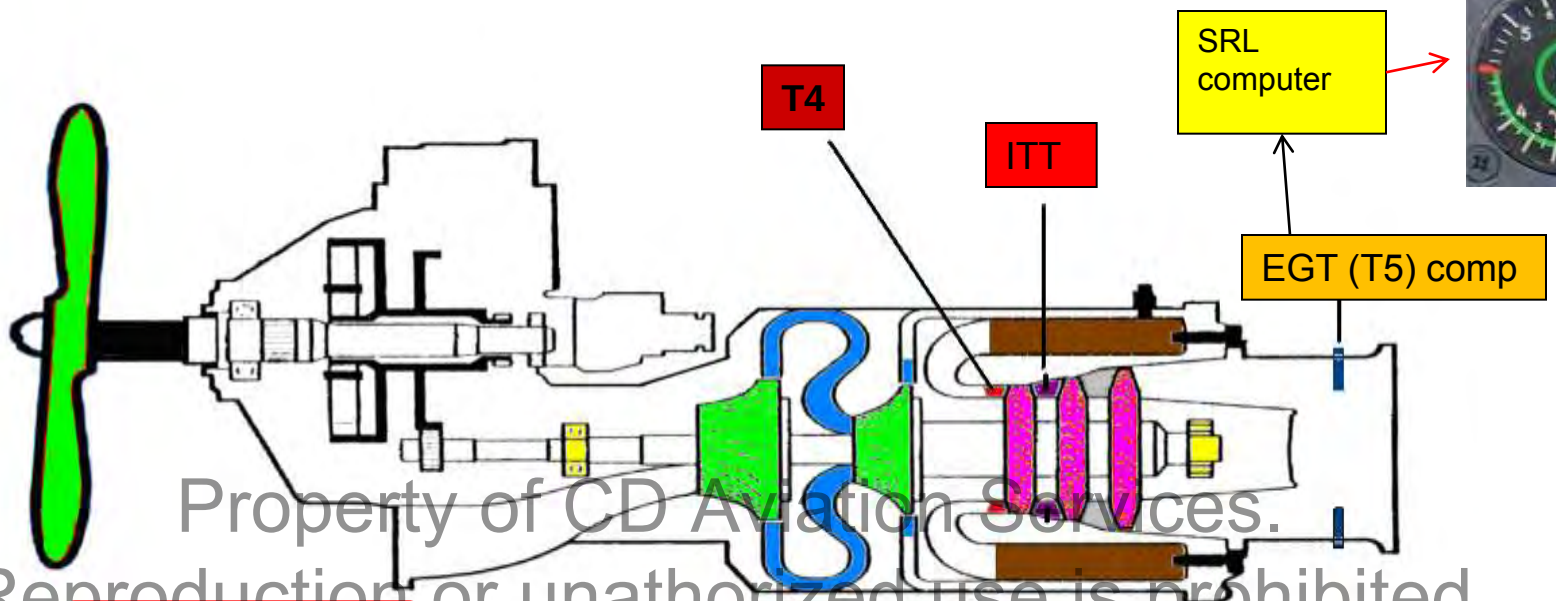
ENGINE EGT/ITT SYSTEMS examples (@ max comp)



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All except Conquest

	T4 / (max–20'c)	comp ITT	comp EGT	-10/ -11/ -12 SRL EGT	SRL correction
Std Day	limit	923	550	650	100
Hot Day	limit	923	580	650	70
Cold Day	limit	923	530	650	120

Engine Power

TPE 331- 5 flat rated to 715 HP Engine

AIRCRAFT INDICATION

OAT	59°f (standard day)
PA	SL
HP	715 aircraft red line
PPH	455
ITT	875°c
(T4)	below the limit

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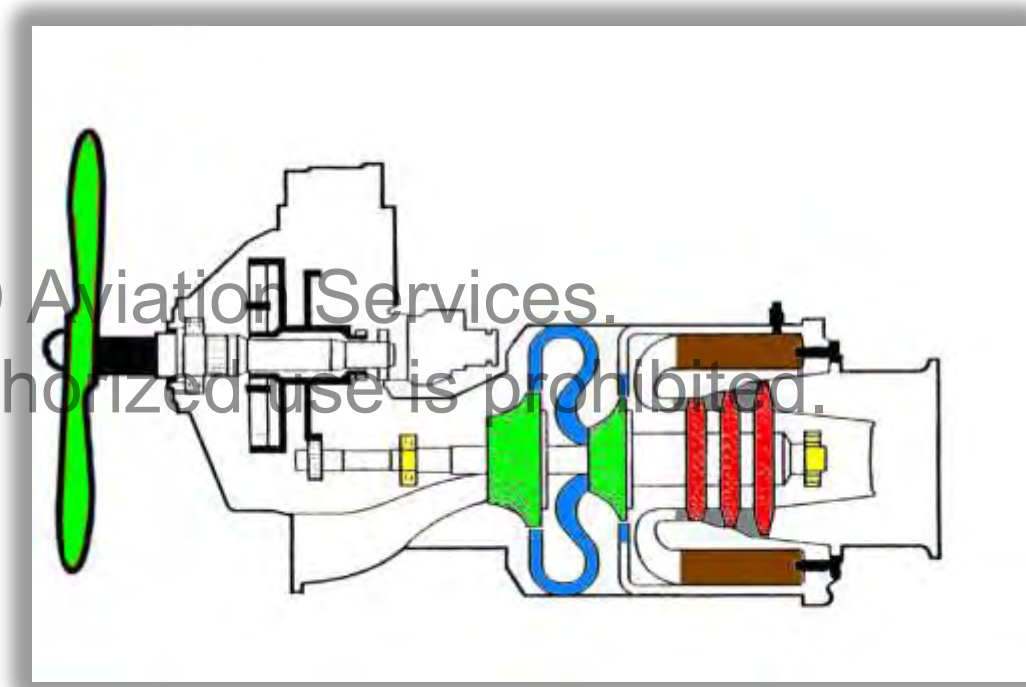
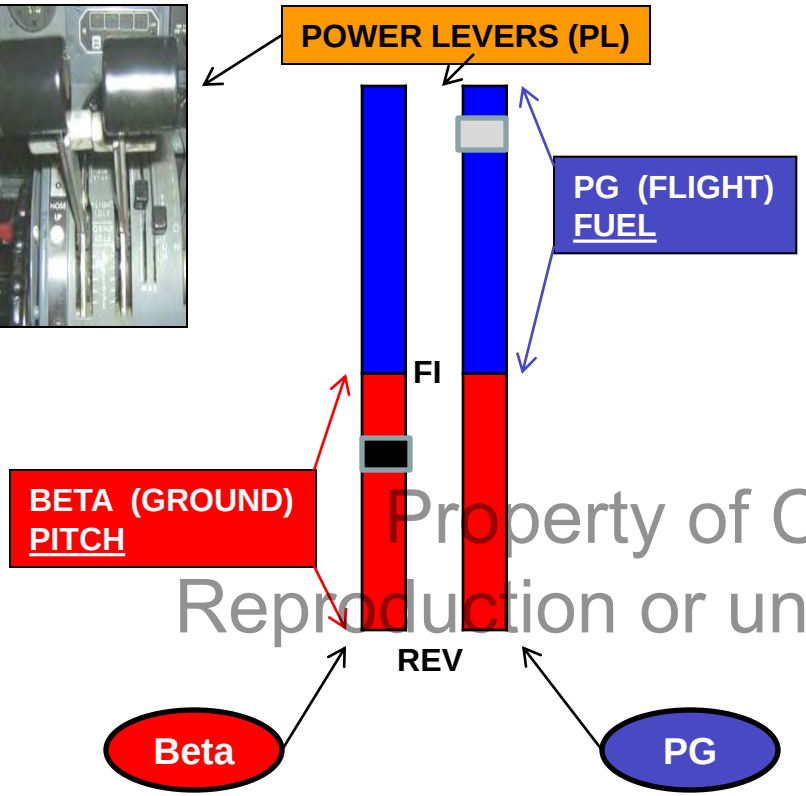
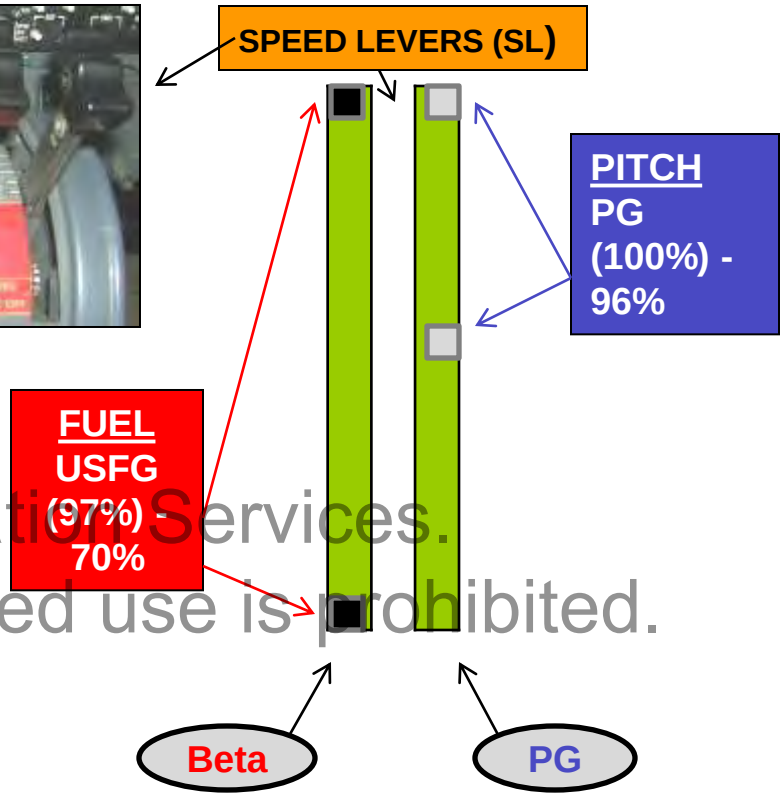


What is the thermal HP capability of this engine?

If 923°c is the ITT limit why did this engine not reach the limit?

Beta and PG Modes of Operation





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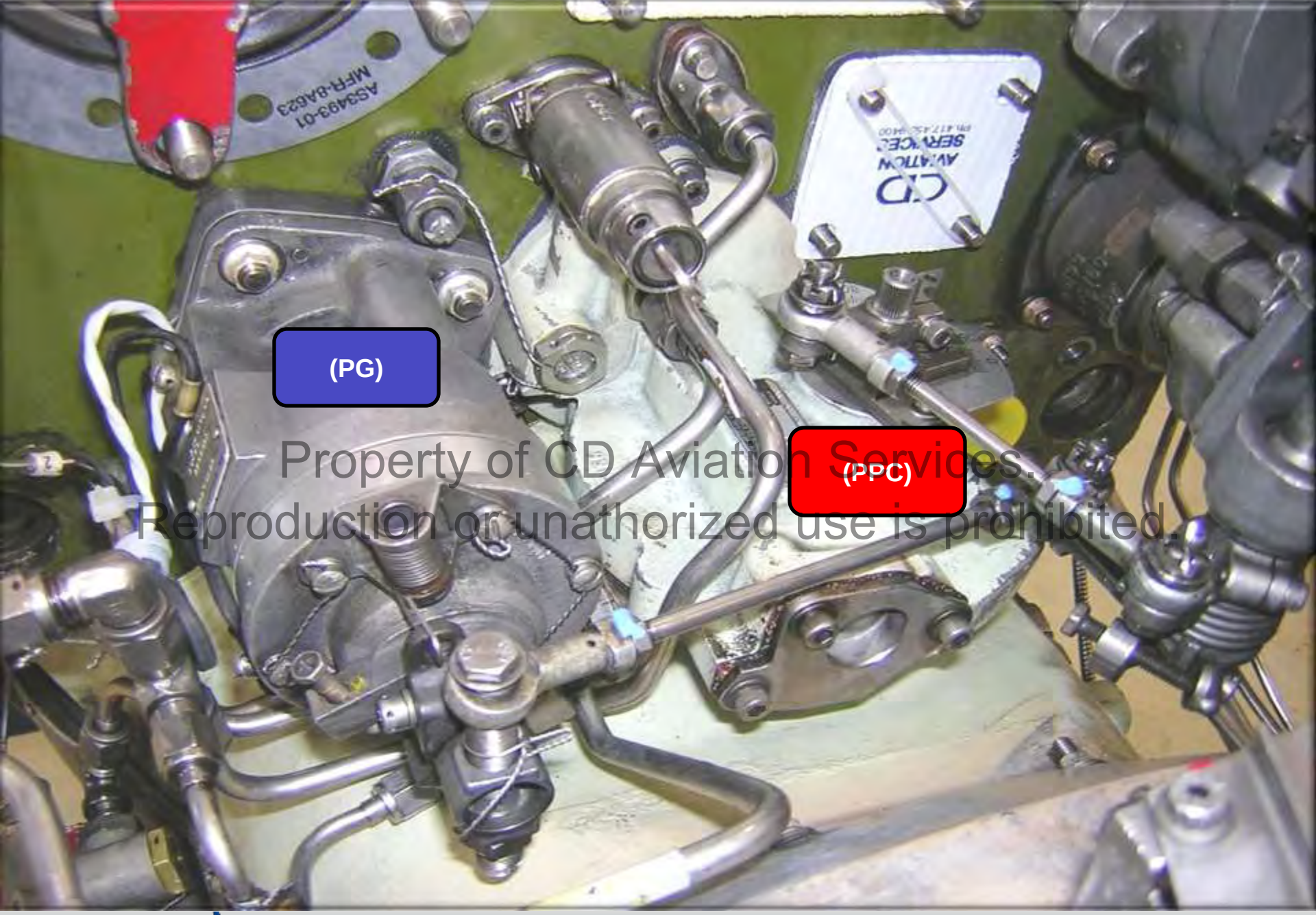
The position of the PL determines the mode of operation
BETA or PG

When the PL is in Beta mode the SL will control the USG speed from 70% to 97%

When the PL is in the PG mode the SL will control the PG speed from 100% to 96%



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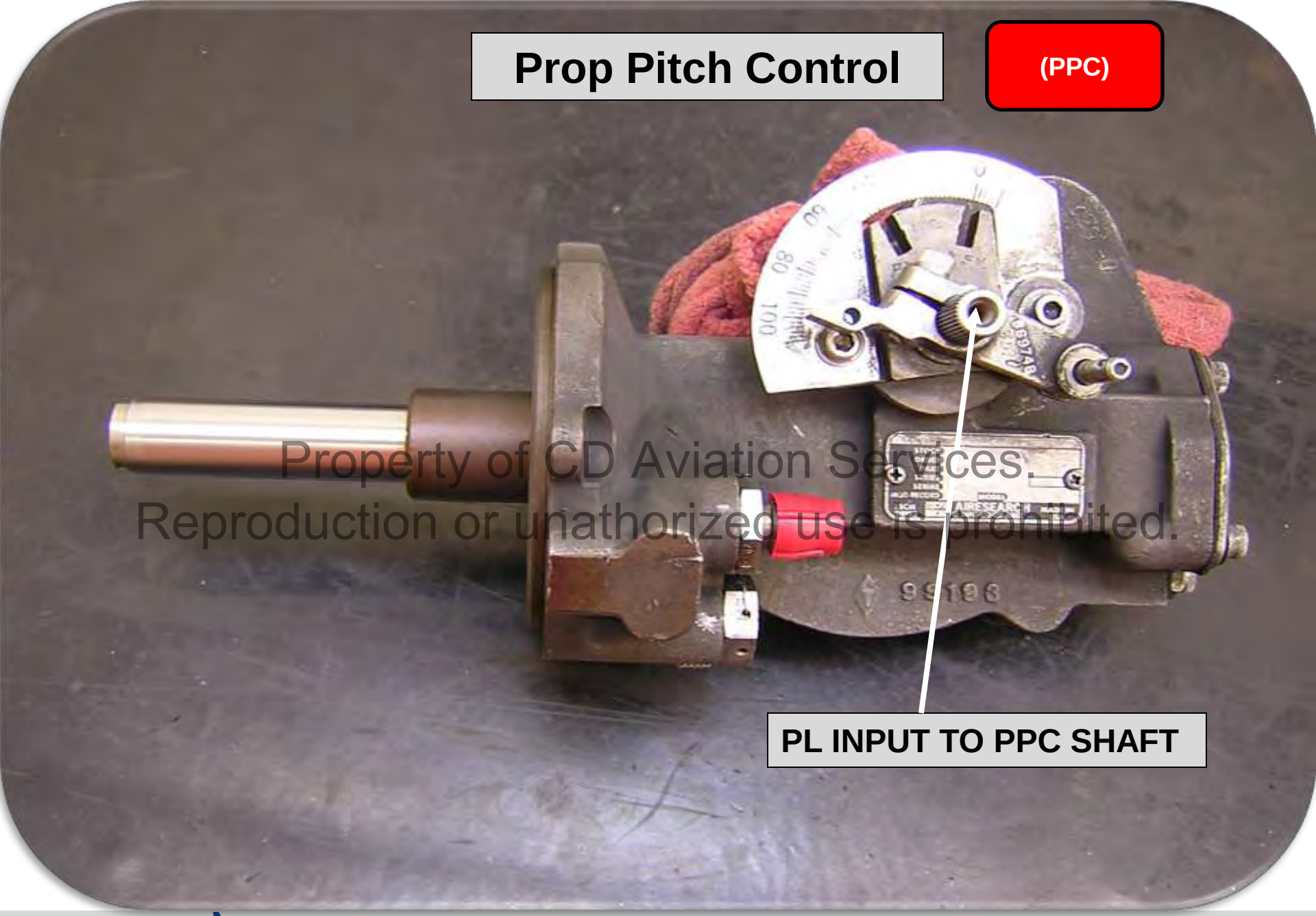
Prop Pitch Control

(PPC)



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Woodward fuel control rigging mock up used during the 1 week line maintenance training class.

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Engine Start and Ground Checks:

What is the min RPM rate of increase?
Why did the NTS light come on and what is the correct RPM for that to occur at?
Is the fuel flow higher than it was on last slide (@ 15% RPM) and why?

NTS
BETA

(T-sensor engines only)

NTS LIGHT OUT GRD CK AND CRICITAL RPM RANGE	
RPM	18-28 %
ITT	400-500°C
HP - PITCH	0
FUEL FLOW	75

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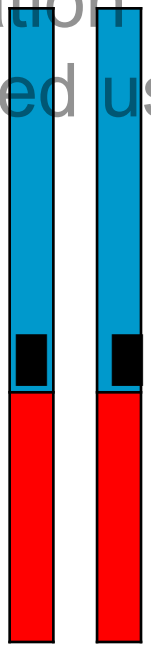
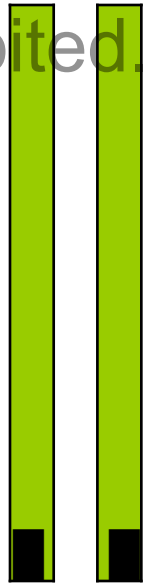
POWER LEVERS (PL)



SPEED LEVERS (SL)



PRIME

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Trouble Shooting

WHAT IS WRONG? WHERE WOULD YOU GO TO ADJUST THIS?

WHAT RPM SHOULD YOU AVOID INFLIGHT IF YOU HAVE A SRL SYSTEM?

DOES THE RH ENGINE NEED MAX FLOW?

PA	1000 feet	
OAT	90°F	
TARGET TORQ	HP	
BETA LIGHT	OFF	OFF
RPM %	96.0	96.0
EGT ° C	550	560
HP	475	485
FF PPH	380	385

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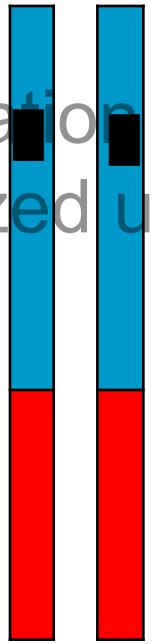
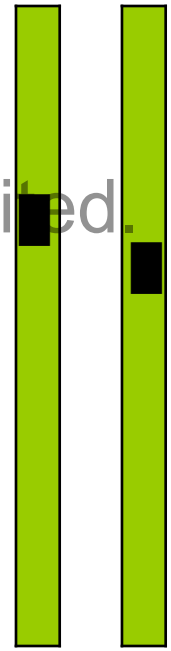
POWER LEVERS PL



SPEED LEVERS SL



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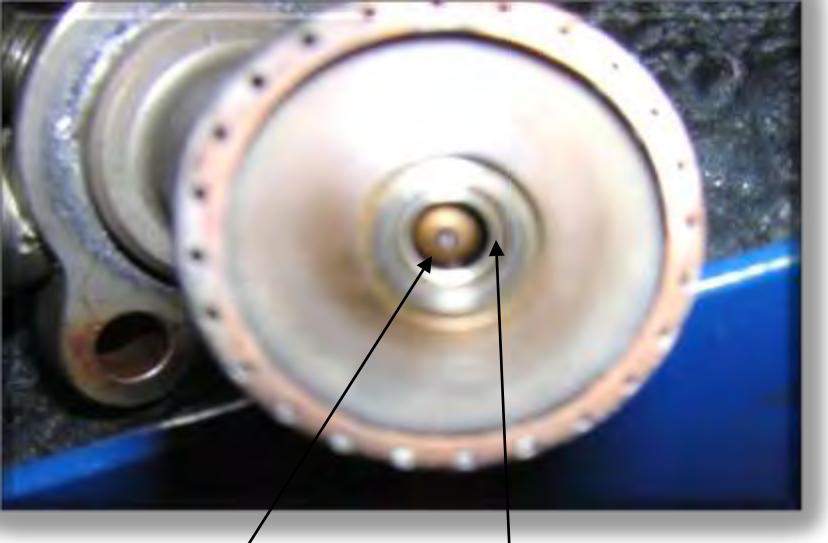
TPE -1 thru -6 engine fuel nozzles

FIVE AND DIME FUEL NOZZLES

DUPLEX FUEL NOZZLES



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FUEL NOZZLE CLEANING INTERVALS

When the engine is running and fuel is flowing through the nozzles the fuel flow cools the nozzle body's and tips below the coking level.

However after engine shut down when the large sections of hot metal in the turbine section radiates heat outward it elevates the temperature of the fuel nozzle body's and tips above the normal operating range and into the range where residual fuel in the fuel nozzle body's and tips may turn into carbon.

Inadequate cooling (3 min at idle required) prior to engine shut down will result in elevated rotating group mass residual temperatures after shutdown. These higher rotating group temperatures will radiate outward and cause elevated fuel nozzle tip and body temperatures a few minutes after engine shut down.

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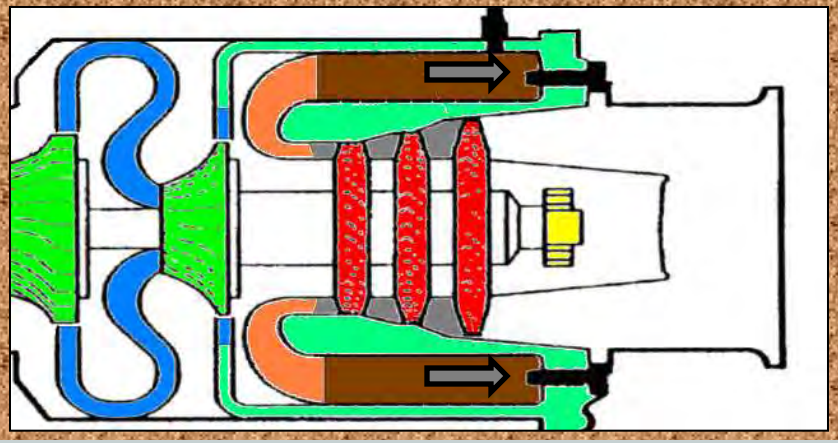
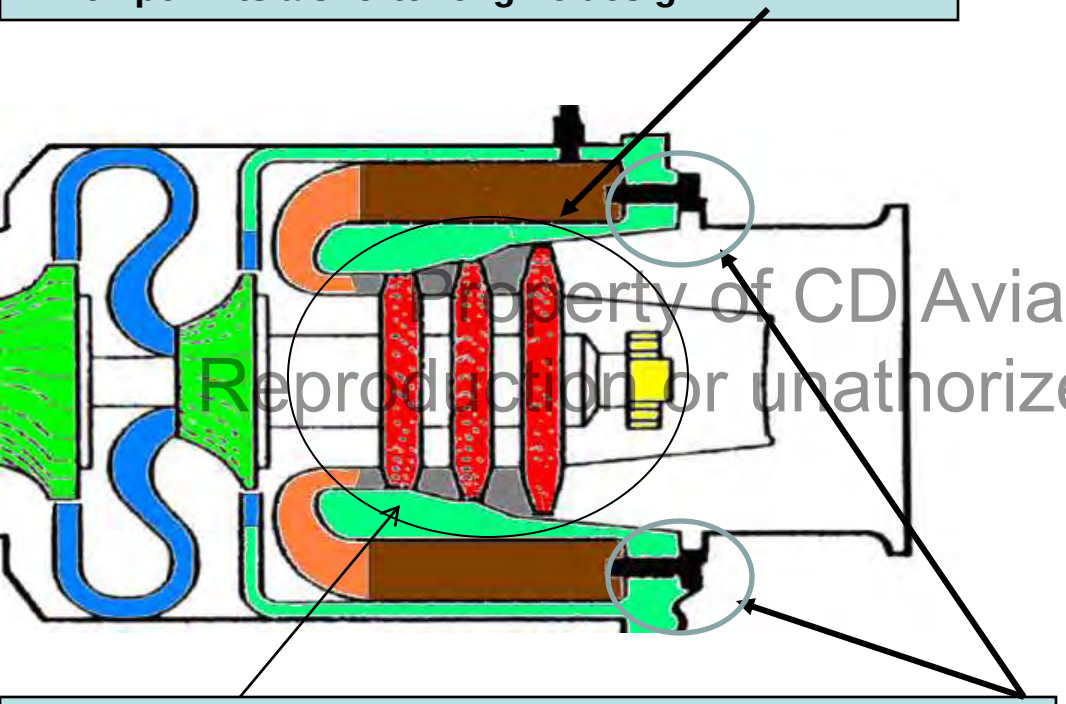
Previous test at HW with improper shut down procedures indicated that fuel nozzle temperatures can be up to 100 degrees higher with the "carbon rate of accumulation" at these higher temperatures being 4 times worse than normal when improper shut downs with less than the (recommended 3 minutes cool down intervals) being used.



FUEL NOZZLE CLEANING INTERVALS

The combustor surrounds the turbine section which permits a shorter engine design.



The turbine section includes a lot of very hot material and after shut down the heat radiates outward and into the fuel nozzles.

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Fuel Nozzle Maintenance

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FUEL NOZZLE CLEANING INTERVALS

A defective fuel nozzle can cause a hot streak within the engine that exceeds the T4 limit at that particular location in the engine.

But because the EGT / ITT harness reads an average of all the probes in the harness and not the highest reading probe, the potential for damage like this is possible if you have a bad flowing fuel nozzle and it will have very little impact on the average EGT reading.

A bad flowing nozzle can cause a small average temperature increase at the EGT harness which often is not adequate to alert you that the fuel nozzle flow is resulting in a hot streak that can badly burn a first stage vane such as the one on the lower right that has **"nozzle burn through".**

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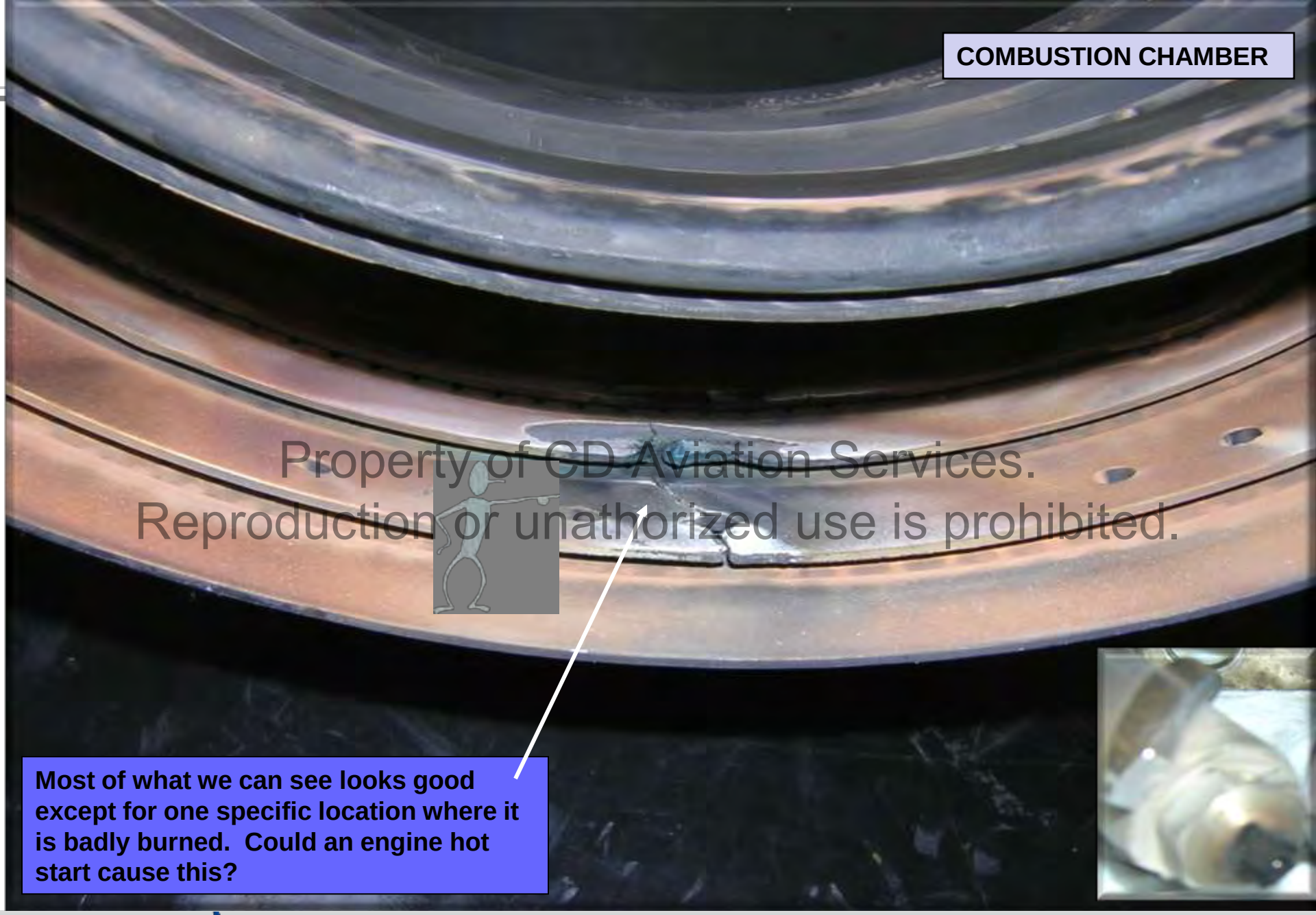
This nozzle vane damage requires that all the first stage turbine blades must be replaced and scrapped due to excess heat cycling and the potential for blade failure if allowed to continued in service.



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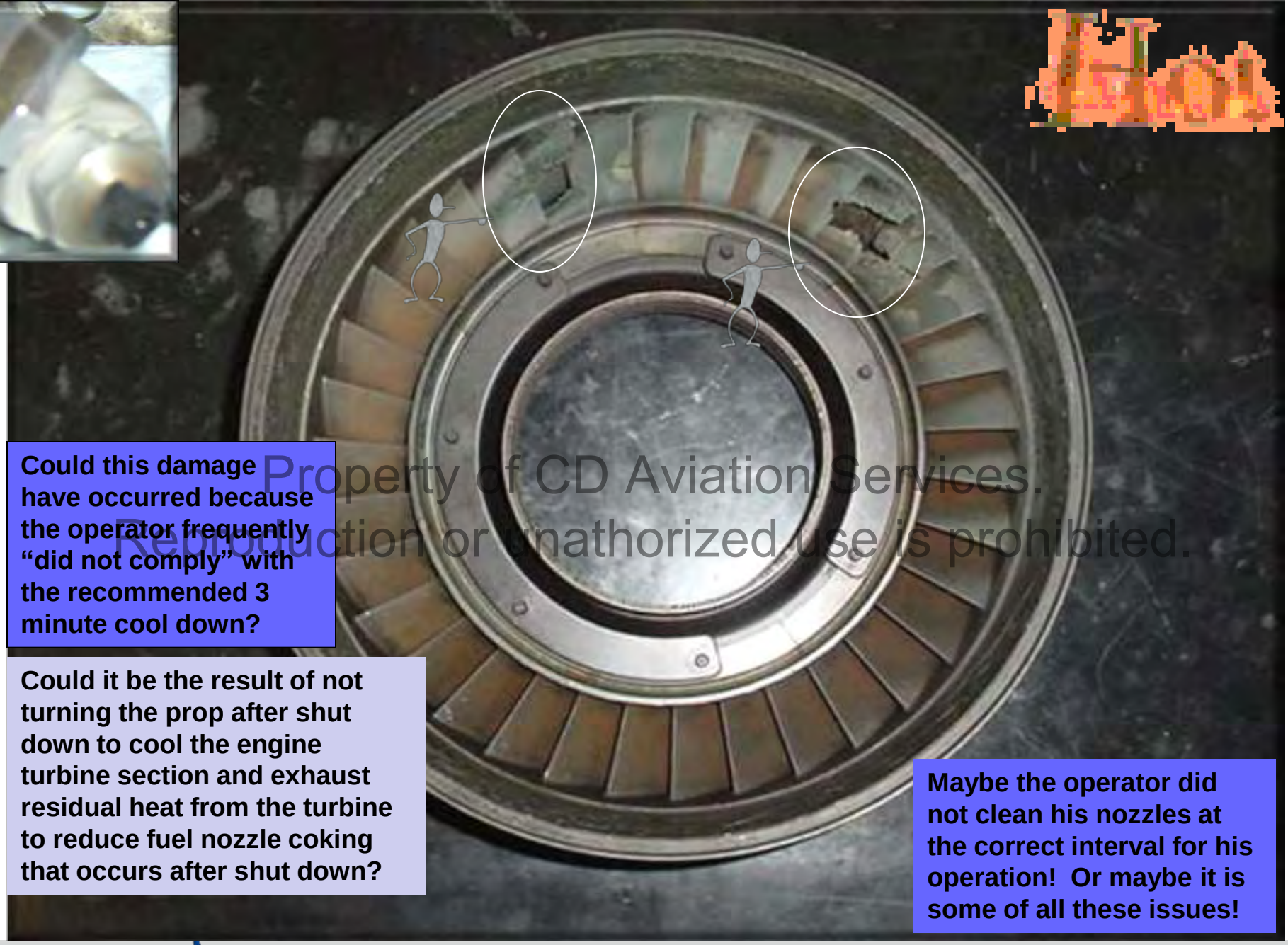

Could this damage have occurred because the operator frequently "did not comply" with the recommended 3 minute cool down?

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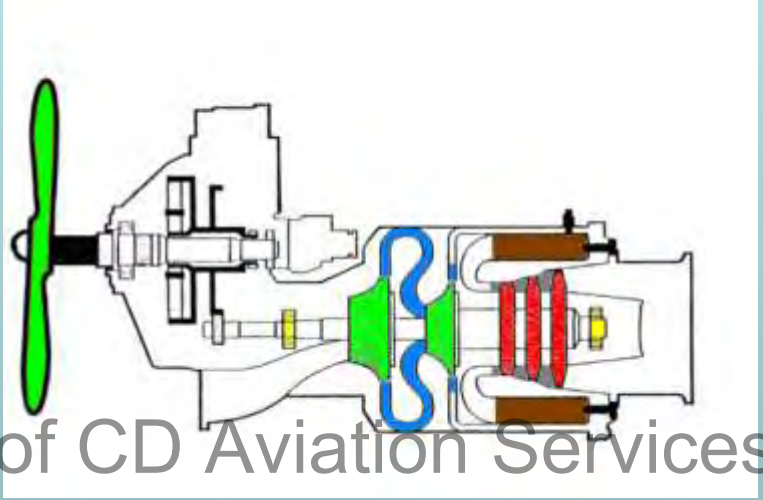
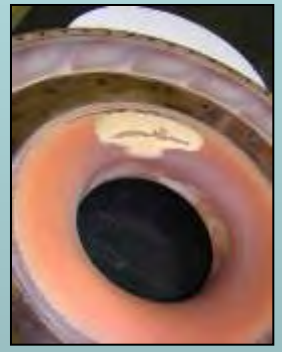
Could it be the result of not turning the prop after shut down to cool the engine turbine section and exhaust residual heat from the turbine to reduce fuel nozzle coking that occurs after shut down?

Maybe the operator did not clean his nozzles at the correct interval for his operation! Or maybe it is some of all these issues!

Operational Cost Savings

The engine <u>critical range</u> is in the 18% to 28% RPM band.





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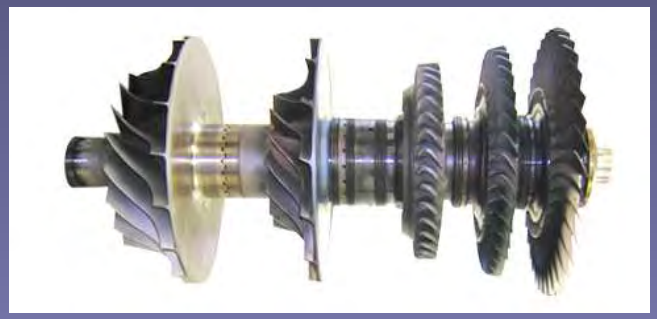
Slow acceleration at a rate of (1% per second or less) will result in the engine main rotating group shaft assembly becoming <u>unstable and flexing</u>. Rotating group flexing radially, will result in the spinning rotating group parts rubbing into the surrounding stationary parts *causing internal engine damage*.

Some of the reasons that an engine may not properly accelerate in the critical range are low fuel flow, low battery assist, weak power cart, power lever in the wrong position, prop not on the start locks and rubs from thermal bow of the main rotating group and etc.

Bowed Rotor Start

Bowed rotor



A condition known as bowed rotor that occurs due to internal thermal differences and (heat rising) within the main rotating group after engine shut could cause engine damage on the next start.
This condition may cause some internal engine rub damage and in some cases it may cause slow acceleration due to the rotating parts making contact with the stationary parts within the engine during engine start.

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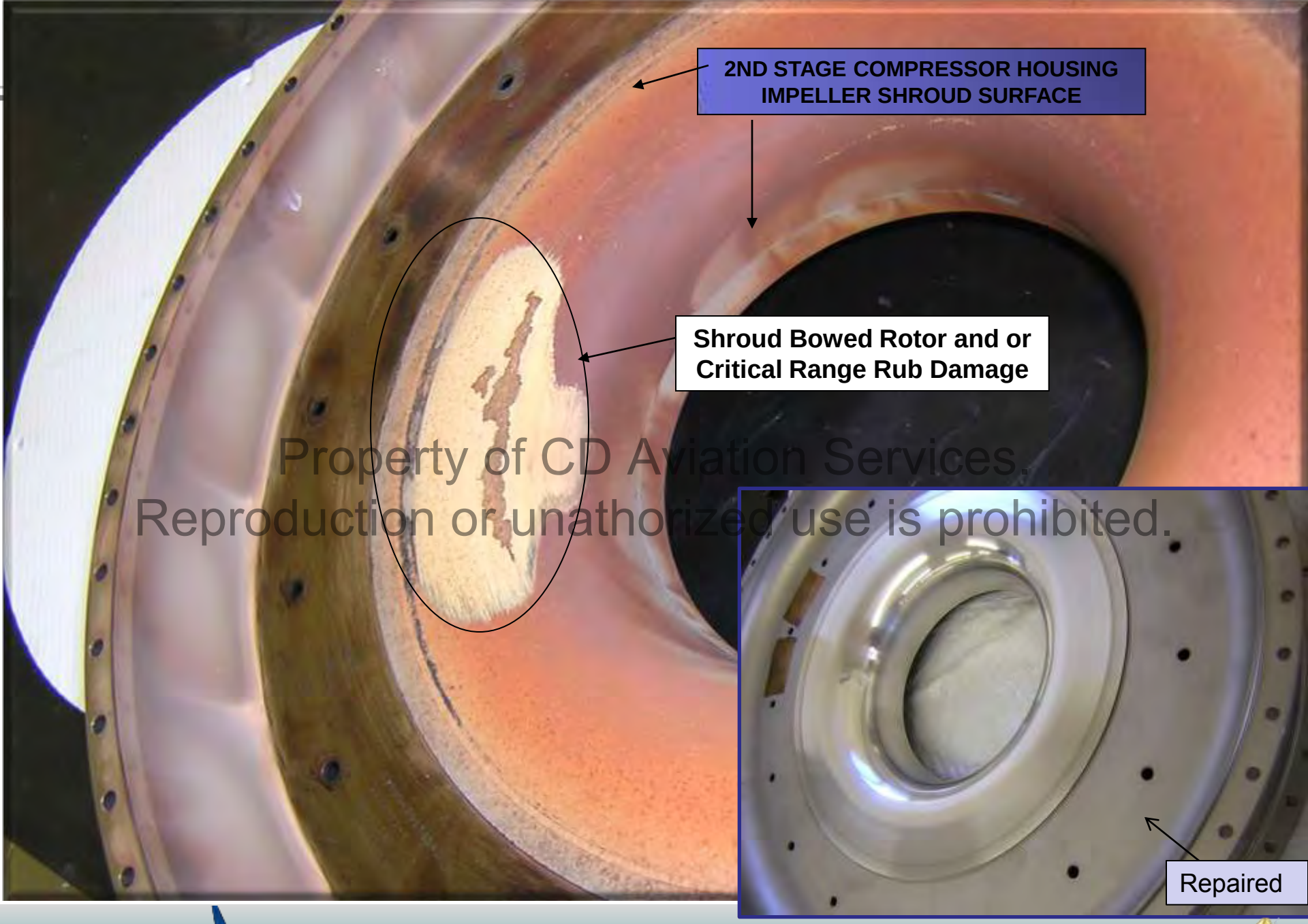
A (slower than normal acceleration rate) in the 18% to 28% (critical range) sometimes referred to as ***hanging in the critical range*** can be due to a bowed rotor causing internal contact and rub. The slow acceleration in critical range can cause engine damage when the main rotating assembly becomes unstable and moves radially into the stationary parts in the critical RPM range.

The bowed rotor condition occurs on all TPE331 engines to some extent but it is more pronounced and more likely to result in hung starts on engines that are freshly refurbished because the internal clearances are tighter on these fresh engines. It is possible on freshly overhauled engines for the engine to totally lock up and not turn 10 to 30 minutes after engine shutdown for 1 hour or more in some cases.



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Operational Cost Savings

Engine operational issues.

Landing with the speed lever low or moving the speed lever from high to low during the landing roll is not recommended for a couple of reasons. If a go around is required the power available with the SL low would not provide adequate power. With SL low inadequate reverse power is available and in full reverse inadequate fuel / RPM may not be available to support the high reverse loads. Even moderate reverse with the SL low especially with high forward airspeeds results in prop loads the engine is not designed for and bog down with damage may occur.

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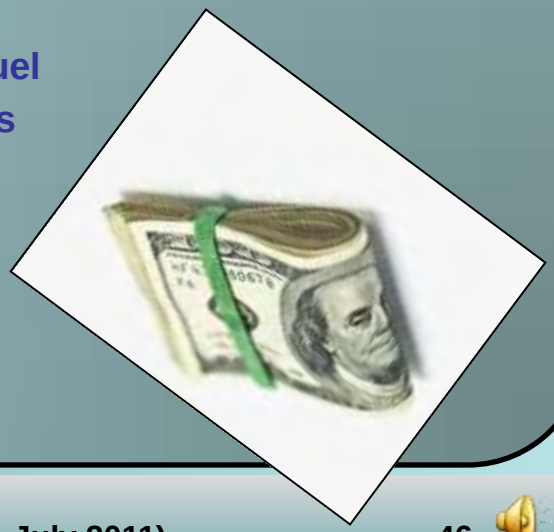
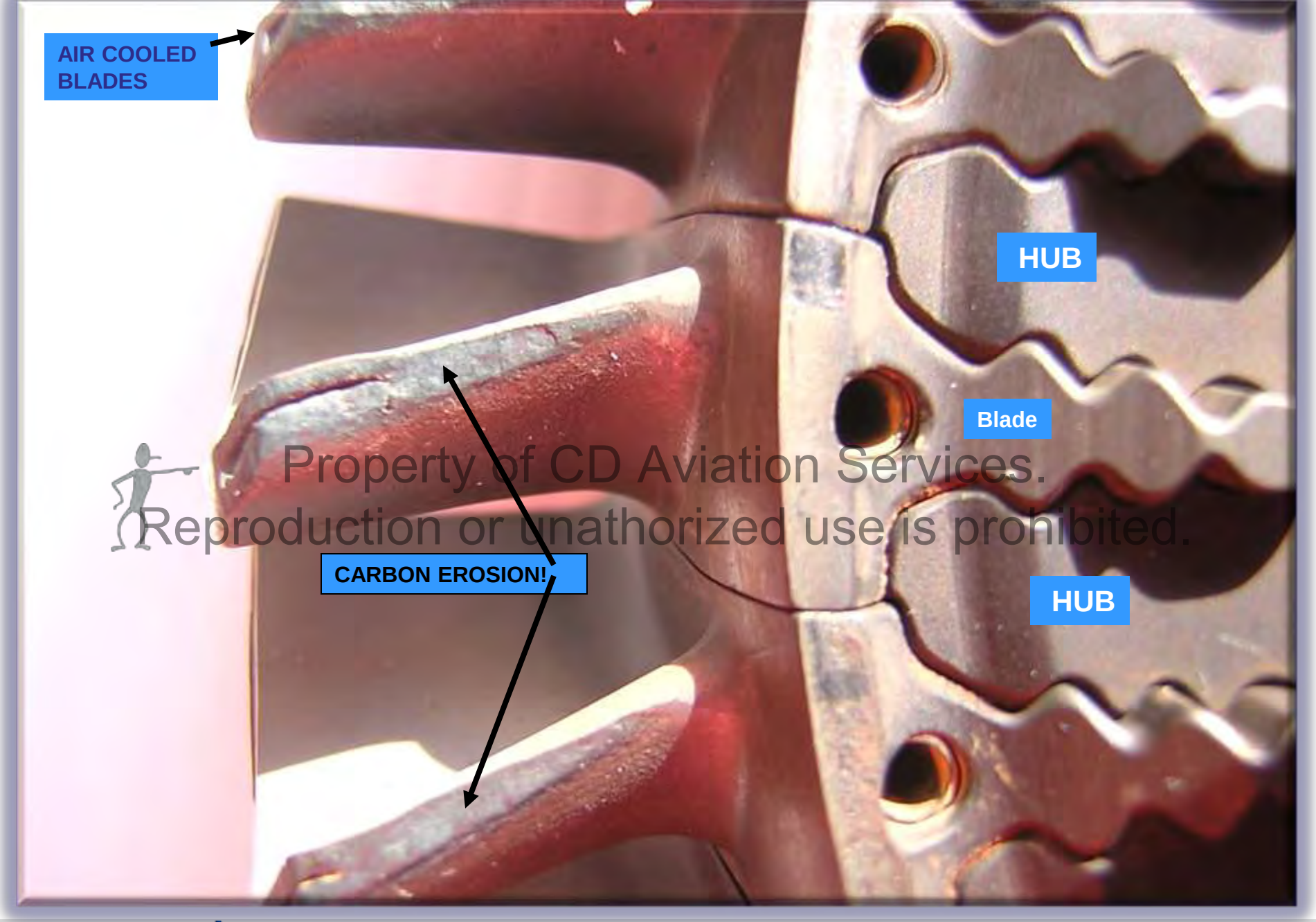
Some engine inlet down AC installations are prone to engine inlet water ingestion during landing on wet runways which can cause engine flame out to occur. The potential flame out is increased further if the SL is retarded to low prior to slowing to taxi speeds.

After a flame out for what ever reason with the accumulating puddle of fuel in the hot turbine section as the engine winds down it may in some cases re-light from residual engine temperatures with the potential for significant engine damage. ues





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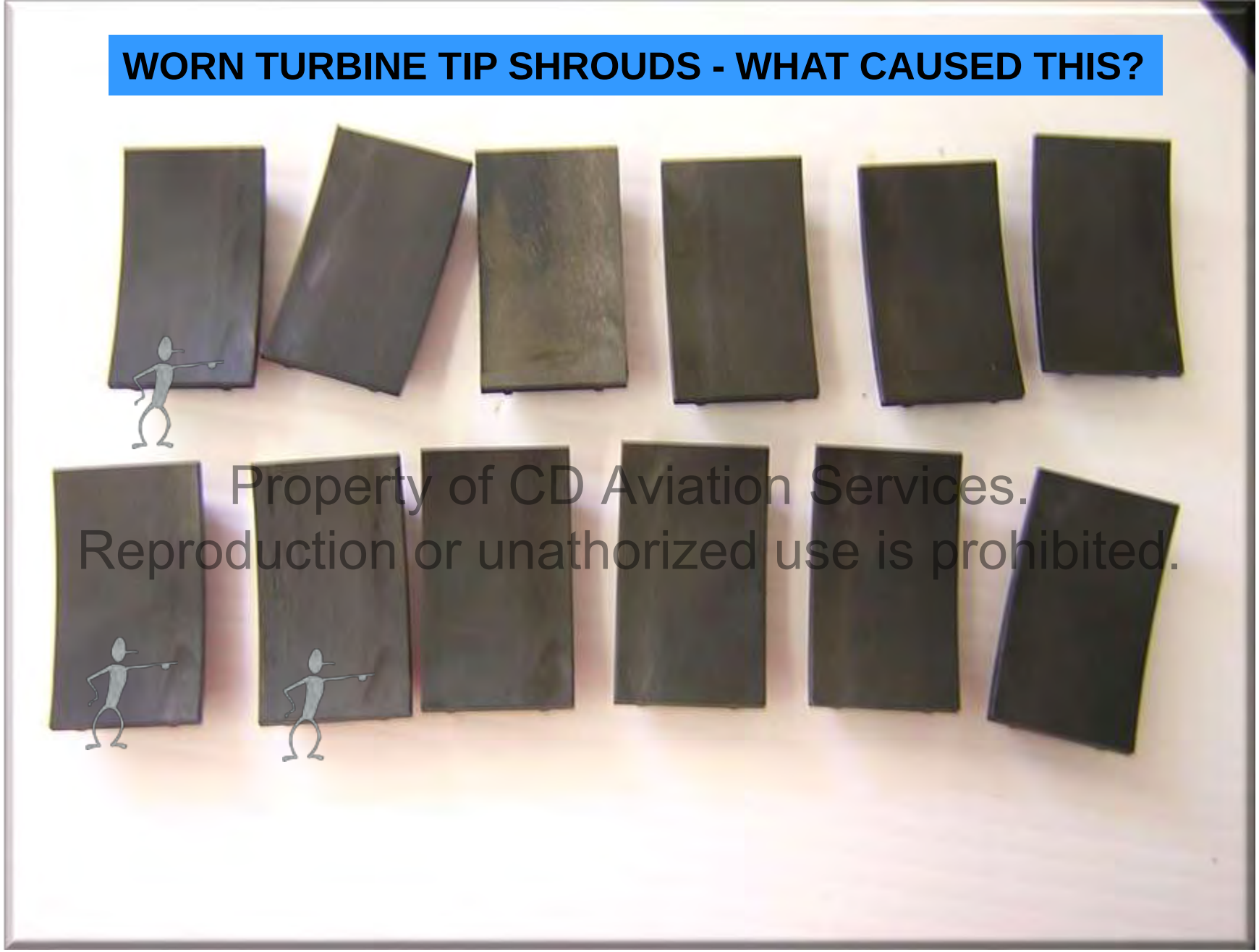
WORN TURBINE TIP SHROUDS - WHAT CAUSED THIS?



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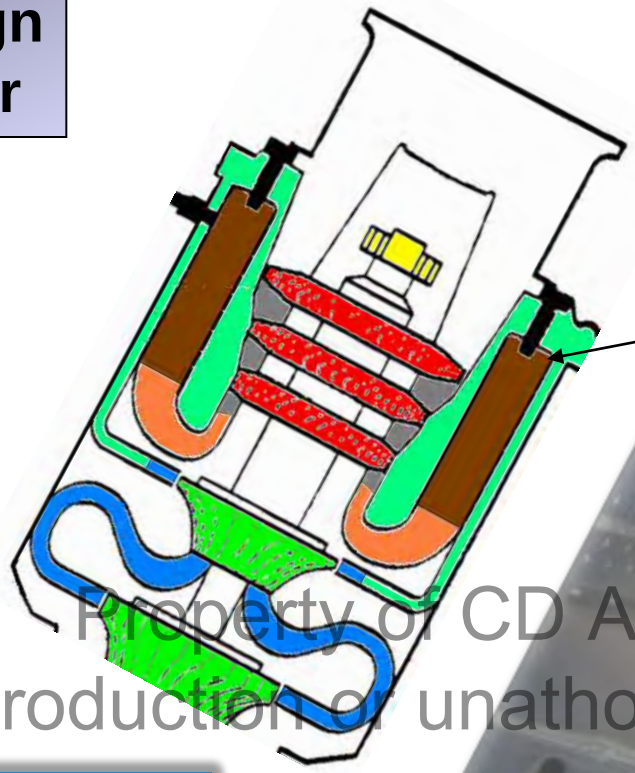
New design combustor



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Main difference is the new design has lots of holes drilled at an angle in aft end of the outer and inner panels

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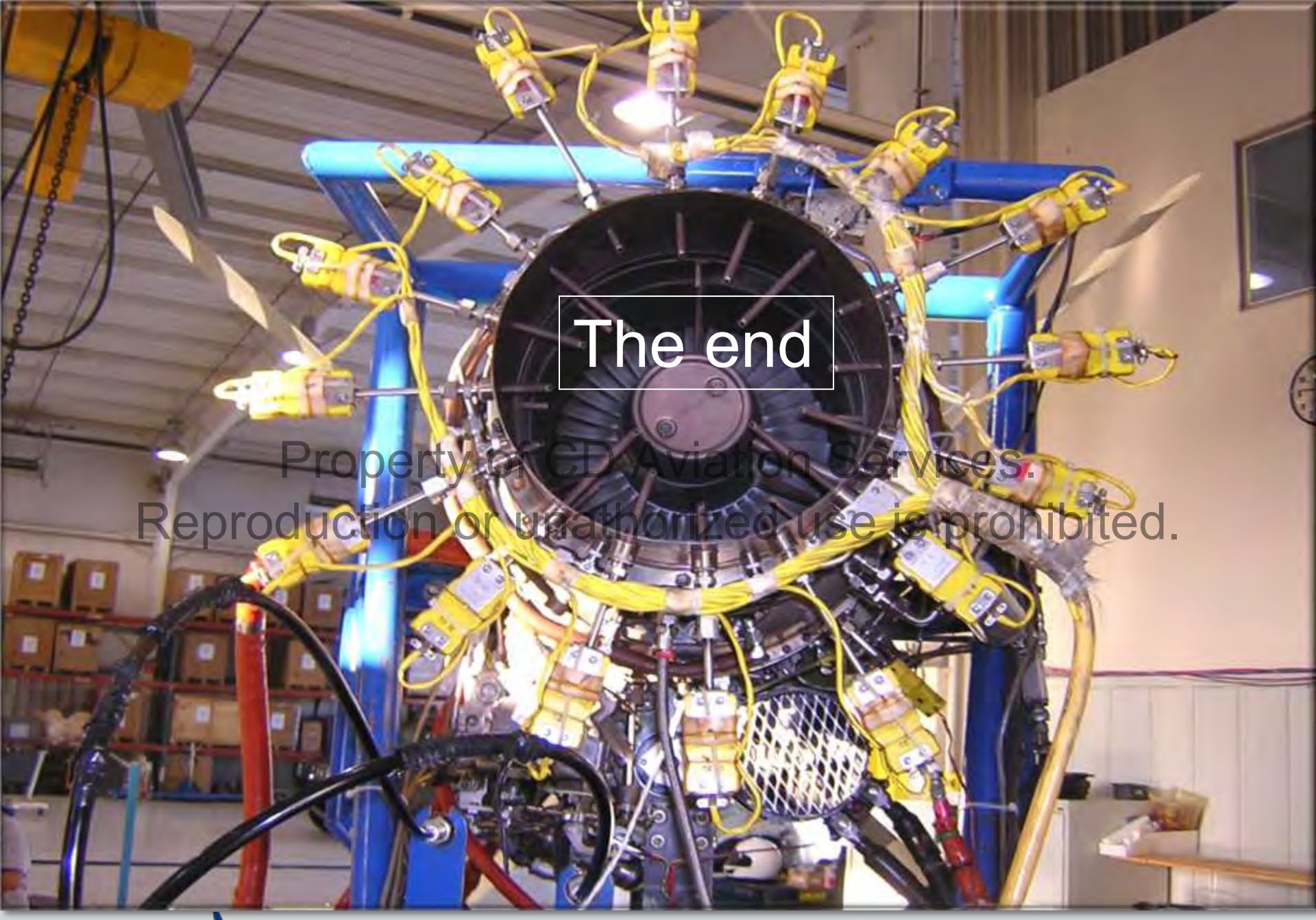
MAIN ROTATING GROUP ASSEMBLY
41730 RPM



The end

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